

THADT: Temporal Hybrid Attention Diffusion Transformer for Human Pose Prediction

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Abstract

Human pose prediction is a key technology for virtual environmental choreography in dance education. However, traditional deterministic prediction methods are unable to capture the diverse distribution of human poses, which limits their practical applicability. Thus, we propose a Temporal Hybrid Attention Diffusion Transformer (THADT) model for 3D to 3D prediction, which consists of forward diffusion and reverse generation processes. During forward diffusion, the discrete cosine transform converts human poses into frequency-domain features while gradually adding noise and training a denoising network to learn the noise distribution. In reverse generation, the model progressively removes noise by integrating historical pose data as conditional input, ultimately reconstructing future pose sequences through inverse transformation. Experimental results on the Human 3.6M and HumanEva-I datasets demonstrate that THADT outperforms existing state-of-the-art methods across key metrics such as ADE, FDE, and MMADE.

CCS Concepts

• **Computing methodologies** → *Activity recognition and understanding.*

Keywords

multimodal prediction, spatiotemporal perception, diffusion

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1 Introduction

Virtual environmental choreography is pivotal for the intelligent and immersive advancement of dance education. Accurate human pose prediction supports this by decoding complex spatiotemporal human movements, thereby enabling precise motion generation for virtual dance [Bahari et al. 2022; Kothari et al. 2021; Saadatnejad et al. 2022]. The objective is to predict future human poses based on historically observed 3D pose sequences. Due to the highly non-linear, temporally dependent, and inherently multimodal nature of human motion, traditional deterministic prediction methods [Chao et al. 2017; Chiu et al. 2019; Fragkiadaki et al. 2015; Ghosh et al.

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2017; Jain et al. 2016; Li et al. 2018; Lucas et al. 2022; Tang et al. 2018] often fail to capture its diverse distribution and are sensitive to observational noise, limiting their practical applicability.

In recent years, probabilistic generative models [Dang et al. 2022a; Hernandez et al. 2019; Lin and Amer 2018; Piergiovanni et al. 2020; Saadatnejad et al. 2023] have emerged as a dominant paradigm for addressing multimodality and uncertainty in human pose prediction. However, existing methods [Dang et al. 2022b; Li et al. 2020; Mao et al. 2019; Xu et al. 2022] still exhibit limitations in long-term temporal dependency modeling, local-global feature integration, and robustness to noise. To address these limitations, the paper proposes a novel Temporal Hybrid Attention Diffusion Transformer (THADT) model. The framework employs a denoising diffusion probabilistic model [Saadatnejad et al. 2023] for progressive generation, enhances spatiotemporal feature extraction with a Temporal Hybrid Attention Module (THAM), and utilizes a Transformer architecture [Lucas et al. 2022] for global context modeling.

The main contributions of the paper are summarized as follows: (1) A diffusion-based generative framework operating in the frequency domain via Discrete Cosine Transform (DCT) to effectively capture periodic motion patterns. (2) The THAM combines self-attention mechanisms with spatial convolutional operations to jointly model long-range dependencies and local features. (3) Extensive experiments on public datasets Human 3.6M and HumanEva-I demonstrate that THADT significantly outperforms existing methods in both prediction accuracy and diversity.

2 Related Work

Human pose prediction represents a significant research direction in computer vision and artificial intelligence, with methodologies evolving from coarse-grained motion analysis to fine-grained joint-level prediction. Early research primarily focused on modeling holistic body movement trajectories, where studies on predicting the temporal evolution of human body centers have established a relatively comprehensive framework [Bahari et al. 2022; Kothari et al. 2021; Saadatnejad et al. 2022]. With advancements in deep learning techniques, research attention has shifted toward fine-grained 3D human joint modeling, forming a spatiotemporal collaborative modeling paradigm that integrates spatial feature extraction with temporal dependency learning for more accurate and coherent predictions.

In recent years, probabilistic generative models have become a dominant paradigm for modeling the inherent multimodality of human motion. Among these approaches, Generative Adversarial Networks (GANs) have shown remarkable capability in producing diverse and realistic human pose sequences. Anelia et al. [Piergiovanni et al. 2020] introduced a framework that explicitly captures temporal dependencies across pose sequences, learning the underlying generative dynamics from observed data distributions to enable multimodal and diverse motion generation during inference. Dang et al. [Dang et al. 2022a] adopted a Gumbel-Softmax sampling mechanism to effectively approximate the multimodal target distribution by sampling from an auxiliary latent space and mapping the samples to Gaussian distributions in the generative domain. Notably, diffusion models have been introduced into the

field of human pose prediction [Saadatnejad et al. 2023] due to their powerful noise modeling and progressive generation capabilities, showing promising potential.

Meanwhile, Transformer-based architectures have attracted extensive attention for their superior capability in capturing global dependencies. Thomas et al. [Lucas et al. 2022] introduced an autoregressive Transformer framework that encodes human motion into discrete latent representations and subsequently decodes them to generate future poses, thereby significantly enhancing the modeling of long-range temporal dependencies. In parallel, Graph Convolutional Networks (GCNs) have demonstrated distinct strengths in capturing the structural topology of human joints. Mao et al. [Mao et al. 2019] developed a GCN that automatically learns inter-joint connectivity while jointly modeling temporal smoothness and spatial dependencies, thereby improving the model's overall representation capability.

Despite notable progress in prior research, several critical challenges remain unresolved. Firstly, extracting informative features from high-dimensional pose data remains a challenging task. Secondly, effective solutions remain limited for missing data caused by occlusion and abrupt motion. Most importantly, current approaches still exhibit significant limitations in modeling complex spatiotemporal dependencies [Dang et al. 2022b; Li et al. 2020; Mao et al. 2019; Xu et al. 2022]. These limitations are particularly reflected in insufficient modeling of long-range temporal dependencies, inadequate fusion of local and global representations, and limited robustness to observational noise.

To address these challenges, this study takes a spatiotemporal feature fusion perspective and aims to develop a novel human pose prediction framework that simultaneously addresses the aforementioned challenges. By integrating advanced generative modeling with hierarchical feature extraction, the proposed framework is designed to enhance predictive performance and robustness in complex scenarios.

3 Methods

3.1 Problem Description

Given the H-frame human pose sequence of historical observations $X^{(1:H)} = [X^{(1)}, X^{(2)}, \dots, X^{(H)}] \in \mathbb{R}^{H \times 3J}$, where $X^{(h)} \in \mathbb{R}^{3J}$ represents the three-dimensional position coordinate vector of the human joint in the h-th frame, and J is the total number of human joints. Our goal is to predict the human pose sequence $X^{(H+1:H+F)} = [X^{(H+1)}, X^{(H+2)}, \dots, X^{(H+F)}] \in \mathbb{R}^{F \times 3J}$ of the future F frame based on the given historical three-dimensional human pose sequence $X^{(1:H)} \in \mathbb{R}^{H \times 3J}$.

3.2 Model Framework

The paper proposes the THADT for 3D to 3D human pose prediction, and its overall architecture is illustrated in Figure 1. The THADT integrates the generative capability of diffusion models with the global temporal modeling strength of Transformers, comprising two core stages: a forward diffusion process and a reverse generation process. During the forward diffusion phase, THADT first applies the DCT to transform the input human pose sequence

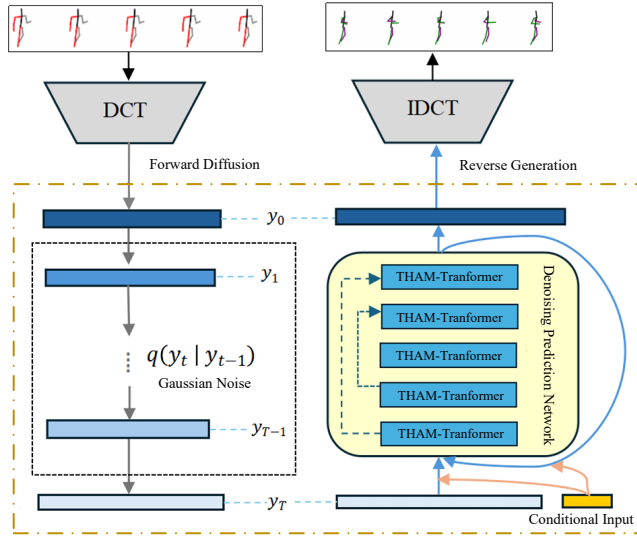


Figure 1: The structure of the Temporal Hybrid Attention Diffusion Transformer (THADT) model

into its frequency-domain representation, thereby isolating informative motion features for subsequent generative modeling. During the t -step forward diffusion process, the model gradually injects Gaussian noise into the frequency-domain representation to approximate a standard normal distribution, while simultaneously training a denoising network to learn the underlying noise distribution. To further strengthen temporal dependency modeling, a THAM is incorporated into the denoising network. This module enables fine-grained extraction of local human pose features and effective modeling of global temporal dependencies, thereby enhancing the model's representational capability for complex human motion patterns. During the reverse generation process, the THADT leverages aggregated diffusion-step encodings together with historical human pose data as conditional inputs to guide the denoising network in progressively removing noise and reconstructing a clean frequency-domain representation. Finally, the THADT converts the denoised frequency-domain representation back into the spatial coordinate domain using the Inverse Discrete Cosine Transform (IDCT), thereby generating the predicted future human pose sequence.

3.3 DCT and IDCT transformation Module

The paper follows the data transformation method in reference [Mao et al. 2019] and converts the human pose sequence X into the frequency-domain representation x through DCT. DCT can remove the high-frequency components irrelevant to human movement by decomposing the human posture frequency domain signal into components of different frequencies while retaining the low-frequency components in the human posture frequency domain signal that represent the human movement pattern. The DCT operation can extract periodic temporal features from human posture data, facilitating the prediction of continuous human pose sequences [Tian et al. 2024]. The process of projecting the human pose frequency-domain

representation x into the DCT domain is expressed as follows:

$$y = DCT(x) = Dx \quad (1)$$

where $y \in \mathbb{R}^{(H+F) \times 3J}$ represents the transformed DCT coefficients, $D \in \mathbb{R}^{(H+F) \times (H+F)}$ denotes the predefined DCT matrix, and the calculation of matrix D is expressed as :

$$D_{i,j} = \sqrt{\frac{2}{(H+F)}} \frac{1}{\sqrt{1+\delta_{i,0}}} \cos\left(\frac{\pi}{2(H+F)}(2j+1)\right) \quad (2)$$

where $\delta_{i,j}$ represents the Kronecker delta, and its calculation method is expressed as:

$$\delta_{i,j} = \begin{cases} 1 & (\text{if } i = j) \\ 0 & (\text{if } i \neq j) \end{cases} \quad (3)$$

Since the DCT operation is an orthogonal transformation, the human pose sequence can be restored from the DCT domain through the IDCT, defined as:

$$x = IDCT(y) = D^T y \quad (4)$$

where D^T denotes the transpose of D . Leveraging DCT and IDCT, the proposed THADT can efficiently process and analyze human pose sequences in the DCT domain. Considering that crucial information about human movement is mainly concentrated in the lower frequency coefficients, it selects the first L rows of D and D^T to simplify the calculation, which are respectively denoted as $D_L \in \mathbb{R}^{(H+F) \times L}$ and $D_L^T \in \mathbb{R}^{(H+F) \times L}$. Then, the noisy DCT coefficients y_t at any diffusion step t is sampled via a reparameterization technique and defined as:

$$y_t = \sqrt{\alpha_t} y_0 + \sqrt{1 - \alpha_t} \varepsilon \quad (5)$$

where $\varepsilon \in N(0, I)$ and $t \in (1, 2, \dots, T)$.

3.4 Denoising Prediction Network Module

We provide a detailed description of the Denoising Prediction Network Module within the THADT, the overall architecture of which is shown in Figure 2. To enhance the model's overall prediction performance, the Denoising Prediction Network Module is constructed by cascading multiple THAM-Transformer blocks. It achieves a prediction of the noise distribution for the input data through temporal modeling and hierarchical feature fusion strategies. As shown in Figure 2, the token sequence composed of historical diffusion step information and human pose data is jointly encoded through an embedding layer to form the initial state representation. This state representation is then fed into the THAM-Transformer block for further feature extraction. Each block consists of three core components. (1) Temporal Hybrid Attention Module (THAM) employs multi-head self-attention to capture long-range temporal dependencies in the data and incorporates spatial convolutions to enhance local feature extraction. (2) Layer Normalization Module, which adopts a pre-activation mechanism to dynamically calibrate the feature distribution, ensuring that the input to each layer remains stable during training, thereby accelerating convergence and improving model robustness. (3) Residual Feedforward Network, comprising linear transformation layers, Swish activation function layers, and Dropout regularization layers, and leverages nonlinear mapping to boost the model's expressiveness. Cross-layer skip connections are employed between THAM-Transformer blocks to

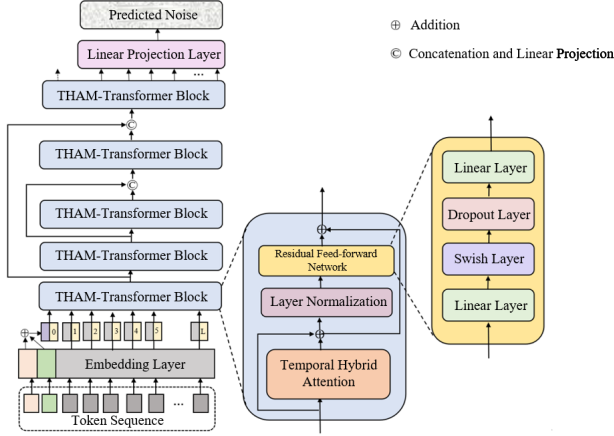


Figure 2: The structure of the Denoising Prediction Network Module

facilitate the fusion of shallow local features with deep global features, alleviating the vanishing gradient problem while improving the model's predictive performance and training stability. Finally, the features processed by a series of THAM-Transformer blocks are fed into a linear projection layer to generate the predicted noise distribution for the input data.

3.5 Temporal Hybrid Attention Module

In human pose prediction, local and global features are crucial for the model to understand the evolution of human pose actions [Li et al. 2022]. We propose a Temporal Hybrid Attention Module (THAM), which employs a hybrid architecture combining self-attention and spatial convolution attention, as shown in Figure 3.

The input feature $F^{(l)}$ is first standardized by a Normalization (Norm) layer and then fed into three fully-connected layers (Linear) for linear transformation to generate the query (Q), key (K), and value (V), respectively. The specific calculation process is defined as:

$$Q = W_Q(\text{LayerNorm}(F^{(l)})) + b_Q \quad (6)$$

$$K = W_K(\text{LayerNorm}(F^{(l)})) + b_K \quad (7)$$

$$V = W_V(\text{LayerNorm}(F^{(l)})) + b_V \quad (8)$$

where W_Q , W_K , and W_V are trainable parameter matrices, while b_Q , b_K , and b_V are bias vectors. Subsequently, the THAM module multiplies the Q with the K and normalizes the result through a Softmax layer to obtain the attention weight distribution. Next, the attention weight distribution is multiplied by V and then added to the input features, yielding the intermediate feature I , which is expressed as:

$$I = \text{Softmax}(QK^T)V + F^{(l)} \quad (9)$$

After obtaining the intermediate feature I , the THAM module extracts local contextual information by integrating an average pooling layer and a 2D convolutional layer, thereby deriving the spatial

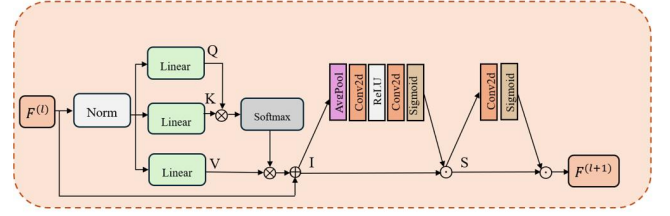


Figure 3: The structure of the Temporal Hybrid Attention Module (THAM)

feature S . The corresponding operation is expressed as:

$$S = I \odot \text{Sigmoid}(\text{Conv2d}(\text{ReLU}(\text{Conv2d}(\text{AvgPool}(I)))) \quad (10)$$

Finally, explicit constraint on feature interaction is achieved through two consecutive Hadamard products, yielding the output feature $F^{(l+1)}$. The complete procedure is described by:

$$F^{(l+1)} = S \odot \text{Sigmoid}(\text{Conv2d}(S)) \quad (11)$$

Ultimately, THAM understands the temporal evolution of human pose actions by integrating the importance of global intermediate features and local spatial features.

4 Experimental

4.1 Datasets

We comprehensively evaluate the proposed method and perform ablation studies on the large-scale dataset Human 3.6M [Ionescu et al. 2013], and show its generalization ability on HumanEva-I [Sigal et al. 2010] datasets. (1) Human 3.6M is the most widely used indoor dataset for the 3D human pose estimation tasks. It consists of 3.6 million images from 4 different cameras, where 11 professional actors perform 15 activities (e.g., eating, smoking) in the indoor environment. The subjects S1, S5, S6, S7, and S8 from the dataset are applied during training, the subjects S9 and S11 are used for testing. (2) HumanEva-I is a much smaller dataset and captures data through three camera views. A single model is trained for all subjects (S1, S2, S3) and all actions (Walk, Jog, Box).

4.2 Implementation Details

The model is implemented using the PyTorch framework with Python 3.9. All experiments are conducted on an Ubuntu 18.04.6 system equipped with CUDA 12.2 and an NVIDIA GeForce RTX 3090 GPU. We employ two benchmark datasets for evaluation: Human 3.6M and HumanEva-I. For the Human 3.6M dataset, the model trains for 1,500 epochs using a denoising network that stacks 8 THAM blocks. The configuration for HumanEva-I differs, with a shorter training schedule of 100 epochs and a shallower network of 4 THAM blocks. During training, each epoch processes 50,000 randomly sampled instances. The diffusion process employs a cosine variance scheduler over 1000 steps. The optimization process uses an initial learning rate of 3×10^{-4} , which we reduce by a factor of 0.8 every 100 epochs.

4.3 Evaluation Metrics

The experimental evaluation indicators of the paper are as follows: (1) Average Pairwise Distance (APD), which calculates the average Euclidean distance between human pose samples and is used to evaluate the diversity of predicted poses. (2) Average Displacement Error (ADE), which calculates the minimum average Euclidean distance between the real attitude sequence and the predicted attitude sequence at all time steps, and is used to evaluate the prediction accuracy. (3) Final Displacement Error (FDE), which calculates the Euclidean distance between the real attitude sequence and the predicted attitude sequence within the last time step, and is also used to evaluate the prediction accuracy of the model. (4) Multi-Modal Average Displacement Error (MMADE), which calculates the average ADE value of all possible future postures generated by the model, and is used to evaluate the average performance of the overall prediction of the model. (5) Multi-Modal Final Displacement Error (MMFDE), which calculates the average FDE value of all possible future postures generated by the model, and is used to evaluate the accuracy of the model in predicting multiple reliable results.

4.4 Comparison with the State-of-the-Art Methods

To evaluate the predictive performance of the proposed model, this study conducts comparative experiments with existing methods on two benchmark datasets: Human 3.6M and HumanEva-I. The experimental results are shown in Table 1. On the Human 3.6M dataset, the proposed THADT achieves values of 0.361, 0.470, and 0.507 for ADE, FDE, and MMADE, which are lower than comparable methods including DeLiGAN[Gurumurthy et al. 2017], HP-GAN[Barsoum et al. 2018], MT-VAE[Yan et al. 2018], MultiObj[Ma et al. 2022], and MotionDiff[Wei et al. 2023].

On the HumanEva-I dataset, the proposed model achieves ADE and FDE values of 0.196 and 0.215, significantly outperforming existing approaches. These results demonstrate its enhanced accuracy in predicting human pose sequences. For the Human 3.6M dataset, our method obtains an APD of 6.135, performing comparably to established baselines including DeLiGAN (6.509) and HP-GAN (7.214), while maintaining competitive sample diversity. Moreover, the model achieves an MMADE of 0.507, surpassing all comparison methods and confirming its capacity to generate multiple plausible predictions with minimal error. Overall, the proposed solution exhibits strong performance in human pose prediction, effectively balancing accuracy and diversity across generated outputs.

4.5 Ablation Study

To verify the effectiveness of the THAM and Skip in the denoising prediction network of the model proposed in the paper, ablation experiments were performed on the HumanEva-I dataset. The experimental results are shown in Table 2. Among them, "Only Skip" indicates that only skip connections are used, and "Only THAM" indicates that only temporal mixed attention modules are used. The first row of Table 2 shows the experimental results using only skip connections, the second row shows the experimental results using only temporal mixed attention, and the third row, "Skip + THAM" presents the experimental results using both skip connections and

temporal mixed attention modules. When solely employing the temporal mixed attention module, the model yields ADE and FDE values of 0.369 and 0.408, respectively, demonstrating significantly higher errors compared to the configuration utilizing both components. Similarly, when only skip connections are implemented, the observed ADE and FDE values reach 0.214 and 0.248, respectively, again exhibiting substantially larger errors relative to the combined approach. These results substantiate the fact that the synergistic integration of temporal mixed attention with skip connections effectively enhances the performance of denoising prediction networks.

To evaluate the impact of the layer depth in the THAM-Transformer block within the denoising prediction network, an ablation study was conducted on the HumanEva-I dataset. The experimental results, summarized in Table 3, demonstrate that an insufficient number of THAM-Transformer layers (e.g., set to 1) leads to model underfitting and fails to effectively capture long-range temporal dependencies. In contrast, when the layer depth is configured to 4, the model achieves optimal prediction performance. These findings validate the effectiveness of the proposed hierarchical feature extraction strategy implemented through a multi-layer architecture.

4.6 Visualization results

To intuitively verify the accuracy of the proposed model, visualization experiments were performed on the HumanEva-I and Human 3.6M datasets, with the results presented in Figures 4(a) and 4(b), respectively. In these figures, the "context" column depicts the preceding frame of the human pose sequence used for prediction, represented by red and black skeletal structures. The "gt" column provides the ground-truth pose sequences, which serve as a reference for evaluating prediction outcomes. The column 'pred #1' illustrates the predicted human pose sequences generated by the model, visualized using green and purple skeletons. A comparative analysis of Figures 4(a) and 4(b) clearly demonstrates that the proposed THADT achieves high predictive accuracy in human pose estimation. In particular, the model not only accurately forecasts human poses over short-term future intervals but also generates multiple plausible and coherent pose sequences, highlighting its robust multimodal predictive capability. These results confirm that the proposed model exhibits strong accuracy and robustness in predicting complex human pose sequences.

5 Conclusion

The paper proposes a 3D human pose prediction model based on a Temporal Hybrid Attention Diffusion Transformer. The model integrates the global modeling advantages of Transformer, the spatio-temporal perception ability of mixed attention, and the progressive generation characteristics of denoising diffusion probability models, significantly enhancing the modeling ability and diversity capture ability of human pose sequences. Systematic experimental analysis and ablation studies on the Human 3.6M and HumanEva-I datasets verified the effectiveness of the model and its various modules. Furthermore, through visualization experiments, the outstanding performance of the model in the paper in the 3D human pose prediction task was demonstrated, further confirming its accuracy and robustness.

Table 1: Experimental results of multiple models on Human 3.6M and HumanEva-I.

Models Indicators	Human 3.6M					HumanEva-I				
	APD	ADE	FDE	MMADE	MMFDE	APD	ADE	FDE	MMADE	MMFDE
DeLiGAN [Gurumurthy et al. 2017]	6.509	0.483	0.534	0.520	0.545	2.177	0.306	0.322	0.385	0.371
HP-GAN [Barsoum et al. 2018]	7.214	0.858	0.867	0.847	0.858	1.139	0.772	0.749	0.776	0.769
MT-VAE [Yan et al. 2018]	0.403	0.457	0.595	0.716	0.883	0.021	0.345	0.403	0.518	0.577
DSF [Yuan and Kitani 2019]	9.330	0.493	0.592	0.550	0.599	4.538	0.273	0.290	0.364	0.340
MOJO [Zhang et al. 2021]	12.579	0.412	0.514	0.497	0.538	4.181	0.234	0.244	0.369	0.347
MultiObj [Ma et al. 2022]	14.240	0.414	0.516	—	—	5.786	0.228	0.236	—	—
GSPS [Mao et al. 2021]	14.757	0.389	0.496	0.476	0.525	5.825	0.233	0.244	0.343	0.331
MotionDiff [Wei et al. 2023]	15.353	0.411	0.509	0.508	0.536	5.931	0.232	0.236	0.352	0.320
THADT [Ours]	6.135	0.361	0.470	0.507	0.540	1.375	0.196	0.215	0.389	0.387

Table 2: Ablation experiments of denoise prediction network modules on HumanEva-I.

Models Indicators	HumanEva-I				
	APD	ADE	FDE	MMADE	MMFDE
Only Skip	1.141	0.214	0.248	0.416	0.436
Only THAM	8.795	0.369	0.408	0.406	0.461
Skip + THAM	1.375	0.196	0.215	0.389	0.387

Table 3: Influence of the number of THAM-Transformer blocks of the denoise prediction network.

Layers Indicators	HumanEva-I				
	APD	ADE	FDE	MMADE	MMFDE
1	7.153	0.362	0.411	0.457	0.461
2	1.736	0.214	0.236	0.380	0.372
3	1.716	0.216	0.237	0.381	0.372
4	1.375	0.196	0.215	0.389	0.387
5	1.371	0.197	0.215	0.389	0.386
6	1.141	0.202	0.225	0.409	0.418
7	1.148	0.201	0.226	0.409	0.420

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**Figure 4: Visualization results on the HumanEva-I and Human 3.6M datasets**

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